PARITY-REDUCED TRANSVERSE WRONSKIANS, JENSEN–LAGUERRE MOMENT IDENTITIES, AND DIVISION-FREE INTERVAL ENCLOSURES FOR RIEMANN'S Ξ-FUNCTION

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ABSTRACT. Let $W(d,t) = \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2$ be the transverse Wronskian of Riemann's completed function $\Xi(t-\mathrm{i}d)$. Because Ξ is real on $d=0$, $W(-d,t) = -W(d,t)$ vanishes to odd order on the critical line, creating a $0/0$ degeneracy in interval enclosures touching $d=0$. Factoring d analytically via $\mathrm{sinc}_{\mathrm{hyp}}(2dy) = \frac{\sinh(2dy)}{2dy}$ yields the canonical parity-reduced representation $\widetilde{W}(d,t) = W(d,t)/d = 32\int_0^\infty y^2\,\mathrm{sinc}_{\mathrm{hyp}}(2dy)I(y,t)\,\mathrm{d}y$, whose Maclaurin coefficients $c_{2m-1}(t) = m\,L_m(\Xi;t)$ recover the classical differential invariants of Jensen (1913) and Csordas–Varga (1990), with $\widetilde{W}(0,t) = L_1(\Xi;t) = (\Xi'(t))^2 - \Xi(t)\Xi''(t)$. Using positive-weight integral representations of $\mathrm{sinc}_{\mathrm{hyp}}$ and η_{hyp}, we construct division-free Cauchy–Riemann gradients and non-harmonic-safe two-dimensional interval mean-value enclosures in `mpmath.iv`. As a methodological benchmark on $t \le 55$, we enclose $c_1, c_3, c_5 > 0$ at $\gamma_1, \dots, \gamma_{10}$, 80 collar boxes around γ_k and true Ξ'-extrema τ_k, and five argument-principle rectangles, while demonstrating why IEEE-754 `float64` quadrature suffers $\mathrm{e}^{+\pi t/4}$ roundoff amplification.

1. INTRODUCTION

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ denote Riemann's completed zeta function, and write $s = \frac{1}{2} + d + \mathrm{i}t$ with transverse offset $d = \sigma - \frac{1}{2} \in \mathbb{R}$ and spectral height $t \in \mathbb{R}$. Setting $z = -\mathrm{i}(s-\frac{1}{2}) = t - \mathrm{i}d$, Riemann's entire function $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z) = R(d,t) + \mathrm{i}I_{\mathrm{im}}(d,t)$ admits the canonical Fourier–Laplace representation [16, 14, 21]

$$\Xi(t-\mathrm{i}d) = 8\int_0^\infty \Phi(u)\cosh(2du)\cos(2tu)\,\mathrm{d}u + \mathrm{i}\,8\int_0^\infty \Phi(u)\sinh(2du)\sin(2tu)\,\mathrm{d}u, \tag{1.1}$$

where the rapidly decaying Jacobi–Riemann kernel $\Phi(u)$ is given by

$$\Phi(u) := \sum_{n=1}^{\infty}\phi_n(u), \qquad \phi_n(u) := \left(2\pi^2 n^4 \mathrm{e}^{9u} - 3\pi n^2 \mathrm{e}^{5u}\right)\exp\left(-\pi n^2 \mathrm{e}^{4u}\right). \tag{1.2}$$

By the equivalence theorems of Hinkkanen [7] and Lagarias [9] (building on Speiser's geometric analysis of $\zeta'(s)$ [17]), the Riemann Hypothesis is equivalent to the positivity of $\mathrm{Re}(\xi'(s)/\xi(s)) = W(d,t)/|\Xi(t-\mathrm{i}d)|^2$ across the half-plane $d > 0$, where the *transverse Wronskian* is given via the Cauchy–Riemann equations ($\partial_d R = \partial_t I_{\mathrm{im}}$, $\partial_d I_{\mathrm{im}} = -\partial_t R$) and the canonical Wigner correlation kernel $I(y,t) := \int_{-\infty}^{\infty}\Phi(\frac{x+y}{2})\Phi(\frac{x-y}{2})\cos(2tx)\,\mathrm{d}x = 2\pi\,\mathcal{W}_\Phi(y/2, 2t)$ of Paper I [11] by

$$W(d,t) := \frac{1}{2}\frac{\partial}{\partial d}\left|\Xi(t-\mathrm{i}d)\right|^2 = R\,\partial_t I_{\mathrm{im}} - I_{\mathrm{im}}\,\partial_t R = 16\int_0^\infty y\sinh(2dy)\,I(y,t)\,\mathrm{d}y. \tag{1.3}$$

When applying two-dimensional interval arithmetic near the critical line $d = 0$—an issue encountered in zero-free box computations for $\zeta'(s)$ and $\xi'(s)$ since Spira [18, 19]—one faces a kinematic *boundary-layer degeneracy*: because $\Xi(t)$ is real-valued on $d = 0$, $I_{\mathrm{im}}(0,t) \equiv 0$ and $\partial_d R(0,t) \equiv 0$, forcing $W(0,t) \equiv 0$ identically for all $t \in \mathbb{R}$. Any naive interval box $\mathbf{B} = [0,\overline{d}]\times[\underline{t},\overline{t}]$ touching $d = 0$ therefore contains zero in $W(\mathbf{B})$.

Analytically dividing $W(d,t)$ by d leads directly to classical differential invariants. On the critical line $d = 0$, the limiting ratio $\lim_{d\to 0} W(d,t)/d = (\Xi'(t))^2 - \Xi(t)\Xi''(t)$ is the classical *Laguerre*

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expression $L_1(\Xi;t)$, while the higher Maclaurin coefficients $c_{2m-1}(t) := \frac{1}{(2m-1)!}\partial_d^{2m-1}W(0,t)$ satisfy $c_{2m-1}(t) = m\,L_m(\Xi;t)$, where $L_m(f;t)$ are the higher Turán–Jensen differential forms introduced by Jensen [8] in 1913 and proved by Csordas and Varga [2] to satisfy $L_m(\Xi;t) > 0$ ($\forall\, m \geq 1$, $\forall\, t \in \mathbb{R}$) if and only if the Riemann Hypothesis holds (see also Griffin, Ono, Rolen, and Zagier [5] for asymptotic hyperbolicity of Jensen polynomials). Likewise, on the numerical side, the zeros of $\zeta(s)$ on $0 < t \leq 55$ were already isolated over a century ago by Gram [4] and extended by Titchmarsh and Comrie [20], Turing [23], and Platt and Trudgian [13] up to height $3 \cdot 10^{12}$.

The purpose of this note is accordingly twofold and strictly methodological: (i) to record the exact canonical Wigner-moment formulation of the parity-reduced Wronskian $\widetilde{W}(d,t) := W(d,t)/d$, its Jensen–Laguerre Taylor coefficients, and division-free Cauchy–Riemann gradients with rigorous non-harmonic-safe Taylor bounds; and (ii) to benchmark directed-rounding interval enclosures (`mpmath.iv`) on $t \leq 55$ while proving quantitatively via Higham's $|x|^\top|y|$ inner-product error bound why fixed-precision IEEE-754 `float64` quadrature breaks down exponentially at moderate heights.

Theorem 1.1 (Canonical Parity-Reduced Wronskian, Jensen–Laguerre Moments, and Interval Enclosures)**.** *Let* $\Xi(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I_{\mathrm{im}}(d,t)$ *be defined by* (1.1)*, and let* $I(y,t) = \int_{-\infty}^{\infty}\Phi(\frac{x+y}{2})\Phi(\frac{x-y}{2})\cos(2tx)\,\mathrm{d}x$ *be the canonical Wigner correlation kernel* [11]*. Then:*

(i) ***Odd parity reduction and canonical hyperbolic-sinc representation:*** *The transverse Wronskian satisfies* $W(-d,t) = -W(d,t)$ *with* $\partial_d^{2k}W(0,t) \equiv 0$ *for all* $k \geq 0$*. The parity-reduced Wronskian* $\widetilde{W}(d,t) := W(d,t)/d$ *extends to an even real-analytic function on* $\mathbb{R}^2$ *given by*

$$\widetilde{W}(d,t) = 32\int_0^\infty y^2\,\mathrm{sinc}_{\mathrm{hyp}}(2dy)\,I(y,t)\,\mathrm{d}y, \qquad \mathrm{sinc}_{\mathrm{hyp}}(w) := \frac{\sinh w}{w} = \int_0^1 \cosh(sw)\,\mathrm{d}s \geq 1, \tag{1.4}$$

with flat Neumann tangent $\partial_d\widetilde{W}(0,t) \equiv 0$ *and boundary value equal to the classical Laguerre expression* $\widetilde{W}(0,t) = L_1(\Xi;t) := (\Xi'(t))^2 - \Xi(t)\Xi''(t)$*.*

(ii) ***Jensen–Csordas–Varga moment identity and local multiplicity jet:*** *The Maclaurin series* $\widetilde{W}(d,t) = \sum_{m=1}^\infty c_{2m-1}(t)\,d^{2m-2}$ *has coefficients equal to* m *times the Jensen–Laguerre invariant* $L_m(\Xi;t)$ [8, 2]*:*

$$c_{2m-1}(t) = \frac{2^{2m+3}}{(2m-1)!}\int_0^\infty y^{2m}I(y,t)\,\mathrm{d}y = m\,L_m(\Xi;t) := m\sum_{j=0}^{2m}\frac{(-1)^{m+j}}{j!\,(2m-j)!}\,\Xi^{(j)}(t)\,\Xi^{(2m-j)}(t). \tag{1.5}$$

At any m*-fold real zero* $t_0 \in \mathbb{R}$ *(*$\Xi(t_0) = \cdots = \Xi^{(m-1)}(t_0) = 0$*,* $\Xi^{(m)}(t_0) \neq 0$*),* $c_1(t_0) = \cdots = c_{2m-3}(t_0) = 0$ *and* $c_{2m-1}(t_0) = \frac{m}{(m!)^2}(\Xi^{(m)}(t_0))^2 > 0$*. Across* $\gamma_1,\ldots,\gamma_{10}$*, directed-rounding interval arithmetic (`mpmath.iv`) encloses* $c_1(\gamma_k), c_3(\gamma_k), c_5(\gamma_k) > 0$*.*

(iii) ***Non-harmonic-safe 2D Interval-MVT enclosures and `float64` cancellation obstruction:*** *Using the positive-weight integral representations of* $\mathrm{sinc}_{\mathrm{hyp}}(w)$ *and* $\eta_{\mathrm{hyp}}(w) := \frac{w\cosh w - \sinh w}{w^3} = \int_0^1\int_0^1 s^2\cosh(s\tau w)\,\mathrm{d}\tau\,\mathrm{d}s$*, all mixed* (d,t) *partial derivatives of the non-harmonic auxiliary integrals in* $\nabla\widetilde{W}$ *are bounded directly by holomorphic strip suprema of* $\Xi^{(m)}$ *without harmonicity assumptions. On the benchmark domain* $t \in [0,55]$*, `mpmath.iv` (`iv.dps = 35`) encloses* $\widetilde{W}(\mathbf{B}_m).a > 0$ *across* 80 *collar boxes spanning* $\gamma_1,\tau_1,\gamma_2,\tau_2,\gamma_3$ *(with true* Ξ'*-roots* $\tau_1 \approx 15.5857$*,* $\tau_2 \approx 22.0980$*) and* $N_{\mathrm{zeros}}(\Omega_k) = 0$ *on five rectangles covering* $[10^{-6}, 1.20] \times [0,55]$*. Conversely, IEEE-754 `float64` quadrature of* $\widehat{W}(d,t) = \mathrm{e}^{\pi t/2}\widetilde{W}(d,t)$ *incurs roundoff error proportional to* $u_{\mathrm{mach}}\mathrm{e}^{+\pi t/4}$*, reaching* 4.51% *relative error at* $t = 55$ *and* 1,375% *breakdown at* $t = 60$*.*

The principal benchmark constants verified in Sections 3–5 are summarized below:

$$\begin{aligned} \widetilde{W}(0,0) &= -\Xi(0)\Xi''(0) \approx 1.1420 \times 10^{-2}, \qquad c_1(\gamma_1) = (\Xi'(\gamma_1))^2 \approx 1.9119 \times 10^{-6}, \\ c_3(\gamma_1) &= 2\,L_2(\Xi;\gamma_1) \approx 2.5881 \times 10^{-7}, \qquad c_5(\gamma_1) = 3\,L_3(\Xi;\gamma_1) \approx 1.1553 \times 10^{-8}, \\ \tau_1 &= 15.58570859\ldots, \quad \tau_2 = 22.09797728\ldots \quad (\Xi'(\tau_k) = 0), \\ \min_{1\le m\le 80} \widetilde{W}(\mathbf{B}_m).a &\approx +1.5592 \times 10^{-12} > 0, \qquad \max_{1\le k\le 5} |N_{\mathrm{zeros}}(\Omega_k)| \le 2.62 \times 10^{-14} < \tfrac{1}{2}. \end{aligned} \tag{1.6}$$

Table 1 places these identities and interval benchmarks in the context of established prior work.

TABLE 1. Relationship between classical prior art and the methodological contributions of this note.

Prior Literature	Established Result	Contribution in This Note
Jensen (1913) [8], Pólya (1927) [15], Csordas–Varga (1990) [2]	Differential invariants $L_m(f;t)$; equivalence RH $\iff$ $L_m(\Xi;t) > 0$ ($\forall\, m \ge 1$, $t \in \mathbb{R}$).	Identification $c_{2m-1}(t) = m\,L_m(\Xi;t) = \frac{2^{2m+3}}{(2m-1)!}\int_0^\infty y^{2m} I(y,t)\,\mathrm{d}y$ and interval enclosures of $c_1, c_3, c_5 > 0$ at $\gamma_1, \ldots, \gamma_{10}$.
Spira (1968–1973) [18, 19], Hinkkanen (1997) [7], Lagarias (1999) [9]	Positivity $\mathrm{Re}(\xi'/\xi) > 0$ on $d > 0$; degeneracy $W(0,t) \equiv 0$ along $d = 0$.	Division-free gradient system $(\widetilde{W}, \partial_t\widetilde{W}, \partial_d\widetilde{W})$ via $\mathrm{sinc}_{\mathrm{hyp}}$ and η_{hyp} with non-harmonic-safe Taylor remainders on $[0,\overline{d}]$.
Gram (1903) [4], Titchmarsh–Comrie (1936) [20], Platt–Trudgian (2021) [13]	Verification of RH up to $T = 3\cdot 10^{12}$ via Gram's law, Turing's method, and FFT interval arithmetic.	Methodological 2D collar + Cauchy winding benchmark on $t \le 55$ resolving true Ξ'-extrema τ_k vs. Z'-extrema.
Higham (2002) [6], Trefethen–Weideman (2014) [22]	Inner-product error $\lvert fl(x^\top y) - x^\top y\rvert \le \gamma_N \lvert x\rvert^\top \lvert y\rvert$ and analytic strip trapezoidal convergence.	Quantitative proof that $\widehat{W}_{\mathrm{float64}}$ suffers $\propto u_{\mathrm{mach}} e^{+\pi t/4}$ error amplification (1,375% error at $t = 60$).

1.1. **Proof overview.** The proof of Theorem 1.1 proceeds in three steps:

(1) *Odd transverse parity and canonical hyperbolic-sinc kernel (Section 2):* From Schwarz reflection $\Xi(t+\mathrm{i}d) = \overline{\Xi(t-\mathrm{i}d)}$, $|\Xi(t-\mathrm{i}d)|^2$ is even in d and $W(d,t)$ is odd in d. Dividing the canonical Wigner integral $W(d,t) = 16\int_0^\infty y\sinh(2dy)I(y,t)\,\mathrm{d}y$ [11] by d and writing $\sinh(2dy)/d = 2y\,\mathrm{sinc}_{\mathrm{hyp}}(2dy)$ yields (1.4) without 0/0 division, valid identically for any real-entire Fourier transform.

(2) *Jensen–Laguerre Maclaurin hierarchy and division-free gradients (Section 3):* Expanding $\mathrm{sinc}_{\mathrm{hyp}}(2dy)$ in powers of $(2dy)^2$ and matching with Leibniz differentiation of $\Xi(t-\mathrm{i}d)\Xi(t+\mathrm{i}d)$ recovers $c_{2m-1}(t) = m\,L_m(\Xi;t)$ (1.5). Differentiating $\mathrm{sinc}_{\mathrm{hyp}}(w)$ via $\mathrm{sinc}'_{\mathrm{hyp}}(w) = w\,\eta_{\mathrm{hyp}}(w)$ yields division-free integral formulas for $\widetilde{W}$, $\partial_t\widetilde{W}$, and $\partial_d\widetilde{W}$.

(3) *Non-harmonic-safe 2D Interval-MVT bounds and Higham cancellation audit (Sections 4–5):* Because the division-free auxiliaries $\widetilde{I} = I_{\mathrm{im}}/d$ and $\widehat{I}_d$ do *not* satisfy Laplace's equation $\partial_d^2 F = -\partial_t^2 F$, mixed (d,t) Taylor remainders cannot be bounded by pure t-derivatives alone. Instead, expressing $\mathrm{sinc}_{\mathrm{hyp}}(w) = \int_0^1 \cosh(sw)\,\mathrm{d}s$ and $\eta_{\mathrm{hyp}}(w) = \int_0^1\int_0^1 s^2\cosh(s\tau w)\,\mathrm{d}\tau\,\mathrm{d}s$ represents every partial derivative $\partial_d^p\partial_t^q F$ as a positive-weight average of the holomorphic derivatives $\Xi^{(m)}(t-\mathrm{i}\delta)$ over $\delta \in [0,\overline{d}]$. We combine these enclosures with Cauchy's argument principle on $t \le 55$ and contrast them against the $e^{+\pi t/4}$ catastrophic cancellation of IEEE-754 `float64` quadrature.

1.2. **Notation and conventions.** Throughout all papers in this series, we maintain a single canonical coordinate and normalization convention: $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$ ($d = \sigma - \frac{1}{2} \in \mathbb{R}$, $t \in \mathbb{R}$), $z = -\mathrm{i}(s - \frac{1}{2}) = t - \mathrm{i}d$, and $\Xi(z) = \xi(\frac{1}{2} + \mathrm{i}z)$ given by (1.1)–(1.2), so that $\Xi(t - \mathrm{i}d) =$

$4\int_{-\infty}^{\infty}\Phi(u)\mathrm{e}^{2(d+\mathrm{i}t)u}\,\mathrm{d}u$ and $\Xi(0)=\xi(\frac12)\approx 0.4971207782$. The canonical mirror variables on $(u_1,u_2)\in\mathbb{R}^2$ are $x=u_1+u_2$ and $y=u_1-u_2$, with Wigner correlation kernel

$$I(y,t):=\int_{-\infty}^{\infty}\Phi\Big(\frac{x+y}{2}\Big)\,\Phi\Big(\frac{x-y}{2}\Big)\cos(2tx)\,\mathrm{d}x=2\pi\,\mathcal{W}_\Phi\Big(\frac{y}{2},2t\Big),\tag{1.7}$$

under which $|\Xi(t-\mathrm{i}d)|^2=16\int_0^\infty\cosh(2dy)I(y,t)\,\mathrm{d}y$, $W(d,t)=16\int_0^\infty y\sinh(2dy)I(y,t)\,\mathrm{d}y$, and $\widetilde{W}(d,t)=32\int_0^\infty y^2\,\mathrm{sinc}_{\mathrm{hyp}}(2dy)I(y,t)\,\mathrm{d}y$. (In terms of the rescaled coordinate $\tilde y=2y$ and $\widetilde\Phi(w)=2\Phi(w/2)$, the prefactor $32y^2\,\mathrm{d}y$ equals $\frac12\tilde y^2\,\mathrm{d}\tilde y$ with $\widetilde I(\tilde y,t)=8I(\tilde y/2,t)$; we work exclusively with the canonical $(y,I(y,t))$ of (1.7) throughout.) We write $\widehat W(d,t):=\mathrm{e}^{\pi t/2}\widetilde W(d,t)$ and $\widehat c_{2m-1}(t):=\mathrm{e}^{\pi t/2}c_{2m-1}(t)$ for the Stirling-envelope-scaled fields. For a real interval $\mathbf{x}=[\underline{x},\overline{x}]$, $\mathbf{x}.a:=\underline{x}$, $\mathbf{x}.b:=\overline{x}$, and $\mathrm{mid}(\mathbf{x}):=\frac12(\underline{x}+\overline{x})$.

2. Odd transverse parity and canonical hyperbolic-sinc representation

We first record the parity reduction of $W(d,t)$ in canonical Wigner coordinates, noting explicitly that this algebraic structure holds for any real-entire function $F(z)$ satisfying Schwarz reflection $F(\bar z)=\overline{F(z)}$ (whether or not F satisfies the Riemann Hypothesis).

Proposition 2.1 (Odd transverse parity and canonical Wigner–hyperbolic identity)**.** *For all $(d,t)\in\mathbb{R}^2$, the transverse Wronskian $W(d,t)=\frac12\partial_d|\Xi(t-\mathrm{i}d)|^2$ is an odd real-analytic function of d,*

$$W(-d,t)=-W(d,t),\qquad \frac{\partial^{2k}W}{\partial d^{2k}}(0,t)\equiv 0\quad(\forall\,k\ge0).\tag{2.1}$$

Consequently, the parity-reduced transverse Wronskian $\widetilde W(d,t):=W(d,t)/d$ (with $\widetilde W(0,t):=\partial_dW(0,t)$) is an even real-analytic function of d satisfying the division-free phase-space identity

$$\boxed{\widetilde W(d,t):=\frac{W(d,t)}{d}=32\int_0^\infty y^2\,\mathrm{sinc}_{\mathrm{hyp}}(2dy)\,I(y,t)\,\mathrm{d}y,\qquad \mathrm{sinc}_{\mathrm{hyp}}(w):=\frac{\sinh w}{w}\ge1.}\tag{2.2}$$

In particular, $\partial_d\widetilde W(0,t)\equiv0$ for all $t\in\mathbb{R}$, and its critical-line value equals the classical Laguerre expression

$$\widetilde W(0,t)=L_1(\Xi;t):=\big(\Xi'(t)\big)^2-\Xi(t)\,\Xi''(t)=32\int_0^\infty y^2\,I(y,t)\,\mathrm{d}y.\tag{2.3}$$

Proof. Since $\Phi(u)$ is real and even on $\mathbb{R}$, (1.1) implies $R(-d,t)=R(d,t)$ and $I_{\mathrm{im}}(-d,t)=-I_{\mathrm{im}}(d,t)$, i.e., $\Xi(t+\mathrm{i}d)=\overline{\Xi(t-\mathrm{i}d)}$. Hence $|\Xi(t-\mathrm{i}d)|^2=R(d,t)^2+I_{\mathrm{im}}(d,t)^2$ is real-analytic and even in d, so $W(d,t)=\frac12\partial_d|\Xi(t-\mathrm{i}d)|^2$ is real-analytic and odd in d, yielding (2.1).

By [11, Theorem 1.1(i)], changing variables $x=u_1+u_2$, $y=u_1-u_2$ (Jacobian $\frac12$) in $|\Xi(t-\mathrm{i}d)|^2=16\iint_{\mathbb{R}^2}\Phi(u_1)\Phi(u_2)\mathrm{e}^{2d(u_1-u_2)}\cos(2t(u_1+u_2))\,\mathrm{d}u_1\,\mathrm{d}u_2$ gives

$$\big|\Xi(t-\mathrm{i}d)\big|^2=16\int_0^\infty\cosh(2dy)\,I(y,t)\,\mathrm{d}y.\tag{2.4}$$

Because $|I(y,t)|\le I(y,0)$ decays super-exponentially ($\ll\mathrm{e}^{8y}\exp(-2\pi\mathrm{e}^{2y})$ as $y\to\infty$), $y^k\sinh(2Dy)|I(y,t)|\in L^1([0,\infty))$ for every $D>0$ and $k\ge0$. Differentiating (2.4) under the integral sign gives $W(d,t)=\frac12\partial_d|\Xi(t-\mathrm{i}d)|^2=16\int_0^\infty y\sinh(2dy)I(y,t)\,\mathrm{d}y$. Dividing by $d\neq0$ and inserting $\frac{\sinh(2dy)}{d}=2y\,\mathrm{sinc}_{\mathrm{hyp}}(2dy)$ yields (2.2). Since $\mathrm{sinc}_{\mathrm{hyp}}(w)=\sum_{m=0}^\infty\frac{w^{2m}}{(2m+1)!}$ is entire and even with $\mathrm{sinc}_{\mathrm{hyp}}(0)=1$, $\widetilde W(d,t)$ is real-analytic and even in d with $\partial_d\widetilde W(0,t)=0$. Finally, at $d=0$, $I_{\mathrm{im}}(0,t)=0$, $\partial_dR(0,t)=0$, $\partial_dI_{\mathrm{im}}(0,t)=-\Xi'(t)$, and $\partial_d^2R(0,t)=-\Xi''(t)$ by Cauchy–Riemann, so $\widetilde W(0,t)=\partial_d(RR_d+I_{\mathrm{im}}I_{\mathrm{im},d})|_{d=0}=(\Xi'(t))^2-\Xi(t)\Xi''(t)$, proving (2.3). □

Remark 2.2 (Classical Laguerre inequality, Lehmer pairs, and off-line counterexamples). Identity (2.3) clarifies both what parity reduction accomplishes and what it does *not* prove unconditionally:

(a) *Laguerre's theorem vs. off-critical zeros:* For any real-entire function F of genus 0 or 1 with *only real zeros* (thus conditionally under RH for Ξ), Laguerre's classical theorem implies $L_1(F;t) = (F'(t))^2 - F(t)F''(t) \geq 0$ for all $t \in \mathbb{R}$, with strict inequality unless t is a multiple zero [8, 15, 1]. Conversely, if F possesses a complex-conjugate pair of off-line zeros $t_* \pm \mathrm{i}d_*$ ($d_* > 0$) near the real axis—as occurs unconditionally for the Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$ [3]—then $L_1(F;t_*)$ can become strictly negative (verified explicitly in Section 6). Thus parity reduction removes the kinematic $0/0$ factor d, whereas global positivity $\widetilde{W}(0,t) > 0$ remains a genuine spectral property.

(b) *Distinction between $\Xi'(t) = 0$ extrema, Hardy's $Z'(t) = 0$ extrema, and Gram points:* Writing $\Xi(t) = -\mathcal{A}(t)Z(t)$ with positive Archimedean envelope $\mathcal{A}(t) = \frac{1}{2}(\frac{1}{4} + t^2)\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{\mathrm{i}t}{2})| \asymp \frac{t^2}{\sqrt{2}}(\frac{2\pi}{t})^{1/4}\mathrm{e}^{-\pi t/4}$, differentiation gives $\Xi'(t) = -[\mathcal{A}'(t)Z(t) + \mathcal{A}(t)Z'(t)]$. Because $\mathcal{A}'(t)/\mathcal{A}(t) \approx -\frac{\pi}{4} + \frac{7}{4t} < 0$ for $t \geq 3$, the critical-line extrema τ_k solving $\Xi'(\tau_k) = 0$ are shifted substantially leftward relative to both the extrema of $Z(t)$ ($t_{Z',1} = 17.88258208\ldots$, $t_{Z',2} = 23.10465065\ldots$) and the Gram points ($g_0 = 17.84559954\ldots$, $g_1 = 23.17028270\ldots$): the first two positive roots of $\Xi'(\tau_k) = 0$ lie at $\tau_1 = 15.58570859\ldots$ and $\tau_2 = 22.09797728\ldots$. At τ_k, $(\Xi'(\tau_k))^2 = 0$ and $\widetilde{W}(0,\tau_k) = -\Xi(\tau_k)\Xi''(\tau_k)$ isolates the local concavity barrier studied by Lehmer [10].

3. Jensen–Laguerre moment hierarchy and division-free Cauchy–Riemann gradients

Expanding $\mathrm{sinc}_{\mathrm{hyp}}(2dy)$ around $d = 0$ identifies the transverse Taylor coefficients of $\widetilde{W}(d,t)$ with the higher differential invariants of Jensen [8] and Csordas–Varga [2].

Theorem 3.1 (Jensen–Csordas–Varga moment hierarchy and local multiplicity jet)**.** *For every $t \in \mathbb{R}$, the parity-reduced Wronskian $\widetilde{W}(d,t)$ expands as the convergent Maclaurin series in d^2,*

$$\widetilde{W}(d,t) = \sum_{m=1}^{\infty} c_{2m-1}(t)\, d^{2m-2} = c_1(t) + c_3(t)\, d^2 + c_5(t)\, d^4 + \mathcal{O}(d^6), \tag{3.1}$$

where for each integer $m \geq 1$, the coefficient $c_{2m-1}(t) := \frac{1}{(2m-1)!}\partial_d^{2m-1} W(0,t)$ satisfies

$$c_{2m-1}(t) = \frac{2^{2m+3}}{(2m-1)!}\int_0^{\infty} y^{2m}\, I(y,t)\,\mathrm{d}y = m\, L_m(\Xi;t) := m\sum_{j=0}^{2m}\frac{(-1)^{m+j}}{j!\,(2m-j)!}\,\Xi^{(j)}(t)\,\Xi^{(2m-j)}(t). \tag{3.2}$$

In particular, the first three transverse coefficients are

$$c_1(t) = L_1(\Xi;t) = \big(\Xi'(t)\big)^2 - \Xi(t)\,\Xi''(t) = 32\int_0^{\infty} y^2\, I(y,t)\,\mathrm{d}y, \tag{3.3}$$

$$c_3(t) = 2\,L_2(\Xi;t) = \frac{1}{6}(\Xi\Xi^{(4)} - 4\Xi'\Xi''' + 3(\Xi'')^2) = \frac{64}{3}\int_0^{\infty} y^4\, I(y,t)\,\mathrm{d}y, \tag{3.4}$$

$$c_5(t) = 3\,L_3(\Xi;t) = \frac{1}{120}(-\Xi\Xi^{(6)} + 6\Xi'\Xi^{(5)} - 15\Xi''\Xi^{(4)} + 10(\Xi''')^2) = \frac{64}{15}\int_0^{\infty} y^6 I(y,t)\,\mathrm{d}y. \tag{3.5}$$

Furthermore, if $t_0 \in \mathbb{R}$ is a real zero of multiplicity $m \geq 1$ ($\Xi(t_0) = \cdots = \Xi^{(m-1)}(t_0) = 0$, $\Xi^{(m)}(t_0) \neq 0$), then $c_1(t_0) = \cdots = c_{2m-3}(t_0) = 0$ and the leading non-zero transverse jet coefficient is the pure square

$$c_{2m-1}(t_0) = \frac{m}{(m!)^2}\Big(\Xi^{(m)}(t_0)\Big)^2 > 0. \tag{3.6}$$

Proof. Inserting $\mathrm{sinc}_{\mathrm{hyp}}(2dy) = \sum_{m=1}^{\infty}\frac{(2dy)^{2m-2}}{(2m-1)!}$ into (2.2) and interchanging summation and integration via Fubini's theorem (justified on $|d| \leq D$ since $\sum_{m=1}^{\infty} 32y^2\frac{(2Dy)^{2m-2}}{(2m-1)!}|I(y,t)| = \frac{16}{D}y\sinh(2Dy)|I(y,t)| \in L^1([0,\infty))$) gives $32\cdot 2^{2m-2} = 2^{2m+3}$, proving the first equality in (3.2).

For the second equality—originally established for general entire functions via Taylor expansions by Jensen [8] and Csordas–Varga [2, Theorem 2.1]—write $|\Xi(t - \mathrm{i}d)|^2 = \Xi(t - \mathrm{i}d)\Xi(t + \mathrm{i}d)$

and apply Leibniz's rule at $d = 0$ using $\partial_d^j \Xi(t - \mathrm{i}d)|_{d=0} = (-\mathrm{i})^j \Xi^{(j)}(t)$ and $\partial_d^{2m-j} \Xi(t + \mathrm{i}d)|_{d=0} = (+\mathrm{i})^{2m-j} \Xi^{(2m-j)}(t)$:

$$\begin{aligned}\frac{\partial^{2m}}{\partial d^{2m}} |\Xi(t - \mathrm{i}d)|^2 \Big|_{d=0} &= \sum_{j=0}^{2m} \binom{2m}{j} (-\mathrm{i})^j \mathrm{i}^{2m-j}\, \Xi^{(j)}(t)\, \Xi^{(2m-j)}(t) \\ &= \sum_{j=0}^{2m} (-1)^{m+j} \binom{2m}{j} \Xi^{(j)}(t)\, \Xi^{(2m-j)}(t).\end{aligned}$$

Since $c_{2m-1}(t) = \frac{1}{(2m-1)!} \partial_d^{2m-1} W(0,t) = \frac{1}{2(2m-1)!} \partial_d^{2m} |\Xi(t - \mathrm{i}d)|^2|_{d=0}$ and $\frac{1}{2(2m-1)!} \binom{2m}{j} = \frac{m}{j!(2m-j)!}$, we obtain $c_{2m-1}(t) = m\, L_m(\Xi; t)$. Evaluating $m = 1, 2, 3$ yields (3.3)–(3.5). Finally, if $\Xi^{(j)}(t_0) = 0$ for $0 \le j \le m - 1$ and $\Xi^{(m)}(t_0) \neq 0$, then for $1 \le n < m$ every summand in $L_n(\Xi; t_0)$ has $\min(j, 2n - j) \le n \le m - 1$ and vanishes, while for $n = m$ only $j = m$ survives with sign $(-1)^{2m} = +1$, giving (3.6). □

We next express $\widetilde{W}(d,t)$ and $\nabla \widetilde{W}(d,t)$ without division by d, together with positive-weight integral representations of $\mathrm{sinc}_{\mathrm{hyp}}$ and η_{hyp} that will be essential for rigorous non-harmonic derivative bounds in Section 4.

Proposition 3.2 (Division-free Cauchy–Riemann gradients and positive-weight kernel integrals)**.** *Let $\Xi(t - \mathrm{i}d) = R(d,t) + \mathrm{i} I_{\mathrm{im}}(d,t)$. Define the even entire kernels*

$$\begin{aligned}\mathrm{sinc}_{\mathrm{hyp}}(w) &:= \frac{\sinh w}{w} = \sum_{k=0}^{\infty} \frac{w^{2k}}{(2k+1)!} = \int_0^1 \cosh(sw)\, \mathrm{d}s \ge 1, \\ \eta_{\mathrm{hyp}}(w) &:= \frac{w \cosh w - \sinh w}{w^3} = \sum_{k=0}^{\infty} \frac{2k+2}{(2k+3)!} w^{2k} = \int_0^1 \int_0^1 s^2 \cosh(s\tau w)\, \mathrm{d}\tau\, \mathrm{d}s \ge \frac{1}{3},\end{aligned} \tag{3.7}$$

and the five division-free Jacobi-theta auxiliary integrals

$$\begin{aligned}\widetilde{I}(d,t) &:= 8 \int_0^\infty \Phi(u)(2u)\, \mathrm{sinc}_{\mathrm{hyp}}(2du) \sin(2tu)\, \mathrm{d}u, \\ \widetilde{I}_t(d,t) &:= 8 \int_0^\infty \Phi(u)(2u)^2\, \mathrm{sinc}_{\mathrm{hyp}}(2du) \cos(2tu)\, \mathrm{d}u, \\ \widetilde{I}_{tt}(d,t) &:= -8 \int_0^\infty \Phi(u)(2u)^3\, \mathrm{sinc}_{\mathrm{hyp}}(2du) \sin(2tu)\, \mathrm{d}u, \\ \widehat{I}_d(d,t) &:= 8 \int_0^\infty \Phi(u)(2u)^3 \eta_{\mathrm{hyp}}(2du) \sin(2tu)\, \mathrm{d}u, \\ \widehat{I}_{td}(d,t) &:= 8 \int_0^\infty \Phi(u)(2u)^4 \eta_{\mathrm{hyp}}(2du) \cos(2tu)\, \mathrm{d}u.\end{aligned} \tag{3.8}$$

Then for all $(d,t) \in \mathbb{R}^2$, $I_{\mathrm{im}}(d,t) = d\, \widetilde{I}(d,t)$, and $\widetilde{W}(d,t)$ and its gradient satisfy

$$\widetilde{W}(d,t) = R(d,t)\, \widetilde{I}_t(d,t) - \widetilde{I}(d,t)\, R_t(d,t), \tag{3.9}$$

$$\frac{\partial \widetilde{W}}{\partial t}(d,t) = R(d,t)\, \widetilde{I}_{tt}(d,t) - \widetilde{I}(d,t)\, R_{tt}(d,t), \tag{3.10}$$

$$\frac{\partial \widetilde{W}}{\partial d}(d,t) = d \Big(\widetilde{I}_t(d,t)^2 - \widetilde{I}(d,t)\, \widetilde{I}_{tt}(d,t) + R(d,t)\, \widehat{I}_{td}(d,t) - \widehat{I}_d(d,t)\, R_t(d,t) \Big). \tag{3.11}$$

Proof. In (3.7), $\int_0^1 \cosh(sw)\, \mathrm{d}s = \sinh(w)/w = \mathrm{sinc}_{\mathrm{hyp}}(w)$. Differentiating with respect to w gives $\mathrm{sinc}'_{\mathrm{hyp}}(w) = \int_0^1 s \sinh(sw)\, \mathrm{d}s = w \int_0^1 s^2\, \mathrm{sinc}_{\mathrm{hyp}}(sw)\, \mathrm{d}s = w \int_0^1 \int_0^1 s^2 \cosh(s\tau w)\, \mathrm{d}\tau\, \mathrm{d}s$, so dividing by w proves the integral identity for $\eta_{\mathrm{hyp}}(w) = \mathrm{sinc}'_{\mathrm{hyp}}(w)/w$. Since $\sinh(2du) = d(2u)\, \mathrm{sinc}_{\mathrm{hyp}}(2du)$, we have $I_{\mathrm{im}} = d\widetilde{I}$, $I_{\mathrm{im},t} = d\widetilde{I}_t$, and $I_{\mathrm{im},tt} = d\widetilde{I}_{tt}$. From $\partial_t W = \partial_t(R I_{\mathrm{im},t} - I_{\mathrm{im}} R_t) = R I_{\mathrm{im},tt} - I_{\mathrm{im}} R_{tt}$, dividing by d yields (3.9) and (3.10). Differentiating $\widetilde{I}$ and $\widetilde{I}_t$ in d via $\frac{\mathrm{d}}{\mathrm{d}w} \mathrm{sinc}_{\mathrm{hyp}}(w) = w\, \eta_{\mathrm{hyp}}(w)$ gives $\partial_d \widetilde{I} = d\, \widehat{I}_d$ and $\partial_d \widetilde{I}_t = d\, \widehat{I}_{td}$. Applying the product rule to (3.9) together with $R_d = I_{\mathrm{im},t} = d\widetilde{I}_t$ and $R_{td} = I_{\mathrm{im},tt} = d\widetilde{I}_{tt}$ yields (3.11). □

4. Directed-rounding interval enclosures and non-harmonic-safe Taylor bounds

To enclose $\widetilde{W}(d,t)$ rigorously over continuum boxes touching $d=0$, an interval algorithm must simultaneously control floating-point roundoff, series and quadrature truncation, and mixed (d,t) Taylor remainders without assuming false harmonicity of the division-free auxiliaries.

4.1. **Dual interval evaluators in `mpmath.iv`.** We implement two independent evaluators in directed-rounding interval arithmetic (`mpmath.iv`, `iv.dps = 35`):

(1) **Evaluator A (Euler–Maclaurin summation):** For $s=\frac{1}{2}+d+it$ and integers $N,M\geq 1$,

$$\zeta(s)=\sum_{n=1}^{N-1} n^{-s}+\frac{N^{1-s}}{s-1}+\frac{1}{2}N^{-s}+\sum_{k=1}^{M}\frac{B_{2k}}{(2k)!}\,(s)_{2k-1}\,N^{-s-2k+1}+R_{M,N}(s), \tag{4.1}$$

with exact rational Bernoulli numbers $B_{2k}\in\mathbb{Q}$ and Backlund–Turing remainder bound [23, 21]

$$|R_{M,N}(s)|\leq \varepsilon_{\mathrm{EM}}(s):=\frac{|s+2M+1|}{\mathrm{Re}(s)+2M+1}\left|\frac{B_{2M+2}}{(2M+2)!}\,(s)_{2M+1}\,N^{-s-2M-1}\right|. \tag{4.2}$$

(2) **Evaluator B (Division-free Jacobi-theta quadrature):** We discretize (1.1) and (3.8) on $[0,U_{\max}]$ ($U_{\max}=1.55$) by the composite trapezoidal rule with $N_u=140$ subintervals ($h=1.55/140\approx 0.0110714$) and Jacobi cutoff $M_{\mathrm{ser}}=12$. Because $\cosh(w)$, $\mathrm{sinc}_{\mathrm{hyp}}(w)$, and $\eta_{\mathrm{hyp}}(w)$ have strictly positive Maclaurin coefficients, their exact interval images on $[\underline{w},\overline{w}]\subset[0,\infty)$ equal $[f(\underline{w}),f(\overline{w})]$ without dependency inflation. By Lemma A.1, on $[0,1.20]\times[0,55]$ the series, tail, and analytic-strip trapezoidal errors satisfy

$$\varepsilon_{\mathrm{ser}}\leq 10^{-220},\quad \varepsilon_{\mathrm{tail}}\leq 10^{-550},\quad \varepsilon_{\mathrm{quad}}\leq 10^{-55},\quad \varepsilon_{\mathrm{trunc}}:=\varepsilon_{\mathrm{ser}}+\varepsilon_{\mathrm{tail}}+\varepsilon_{\mathrm{quad}}\leq 1.01\times 10^{-55}. \tag{4.3}$$

4.2. **Non-harmonic-safe 2D Interval-MVT boxes and Cauchy winding enclosures.** A subtle analytical pitfall arises when enclosing $\nabla\widetilde{W}(\mathbf{B})$ over a rectangle $\mathbf{B}=[\underline{d},\overline{d}]\times[\underline{t},\overline{t}]$: although $R(d,t)=\mathrm{Re}\,\Xi(t-\mathrm{i}d)$ is harmonic ($\partial_d^2 R=-\partial_t^2 R$, so $|\partial_d^p\partial_t^q R|\leq\sup|\Xi^{(p+q)}|$), the division-free auxiliaries $\widetilde{I}(d,t)=I_{\mathrm{im}}(d,t)/d$ and $\widehat{I}_d(d,t)$ are *not* harmonic functions of (d,t). Bounding their mixed (d,t) derivatives naively by pure t-derivatives is therefore invalid. The integral representations in (3.7) resolve this cleanly by expressing $\widetilde{I}$ and $\widehat{I}_d$ as positive-weight averages of genuinely harmonic components across the strip $[0,\overline{d}]$.

Proposition 4.1 (Non-harmonic-safe 2D Interval-MVT and Integer Argument Principle)**.** *Let $\mathbf{B}=[\underline{d},\overline{d}]\times[\underline{t},\overline{t}]\subset[0,0.50]\times[0,55]$ have center (d_0,t_0) and half-widths $\Delta d,\Delta t>0$, and let $M_m:=\sup_{(d,t)\in[0,\overline{d}]\times[\underline{t},\overline{t}]}|\Xi^{(m)}(t-\mathrm{i}d)|$ for $m\geq 0$.*

(a) ***Exact integral majorants for non-harmonic auxiliaries and centered 2D MVT:*** *For every $k\geq 0$ and all integers $p,q\geq 0$, the auxiliary functions $\widetilde{I}^{(k)}:=\partial_t^k\widetilde{I}$ and $\widehat{I}^{(k)}:=\partial_t^k\widehat{I}_d$ satisfy on $\mathbf{B}$:*

$$|\partial_d^p\partial_t^q R^{(k)}|\leq M_{p+q+k},\quad |\partial_d^p\partial_t^q\widetilde{I}^{(k)}|\leq\frac{1}{p+1}M_{p+q+k+1},\quad |\partial_d^p\partial_t^q\widehat{I}^{(k)}|\leq\frac{1}{(p+1)(p+3)}M_{p+q+k+3}. \tag{4.4}$$

Consequently, enclosing each component $F\in\{R,R_t,R_{tt},\widetilde{I},\widetilde{I}_t,\widetilde{I}_{tt},\widehat{I}_d,\widehat{I}_{td}\}$ over $\mathbf{B}$ via its first-order Taylor jet at (d_0,t_0) (using $\partial_d\widehat{I}_d=d\check{I}_{dd}$ with $\zeta_{\mathrm{hyp}}(w):=\eta'_{\mathrm{hyp}}(w)/w$) plus the rigorous second-order remainder $\varepsilon_2^{(F)}:=\frac{1}{2}(M_{2,0}^{(F)}\Delta d^2+2M_{1,1}^{(F)}\Delta d\Delta t+M_{0,2}^{(F)}\Delta t^2)$ from (4.4) encloses $\nabla\widetilde{W}(\mathbf{B})$ via (3.10)–(3.11), and every $(d,t)\in\mathbf{B}$ satisfies

$$\widetilde{W}(d,t)\in\widetilde{W}(\mathbf{B}):=\widetilde{W}(d_0,t_0)+\frac{\partial\widetilde{W}}{\partial d}(\mathbf{B})\cdot[-\Delta d,+\Delta d]+\frac{\partial\widetilde{W}}{\partial t}(\mathbf{B})\cdot[-\Delta t,+\Delta t]. \tag{4.5}$$

(b) ***Integer Argument Principle enclosure:*** *Let* $\Omega = [d_{\min}, d_{\max}] \times [t_{\min}, t_{\max}] \subset (0, 1.20] \times [0, 55]$ *be partitioned counterclockwise along* $\partial\Omega$ *by* $z_0, \dots, z_M = z_0$ ($z_j = t_j - \mathrm{i}d_j$, $h_j = |z_{j+1} - z_j|$, $M_{1,j} \geq \sup_{[z_j, z_{j+1}]} |\Xi'|$). *If* $D_j := R_j R_{j+1} + I_j I_{j+1} > 0$, $\mu_j := \min(|\Xi(z_j)|, |\Xi(z_{j+1})|) - \frac{1}{2} h_j M_{1,j} > 0$, *and* $|\Xi(z_j)| - h_j M_{1,j} > 0$ *for every* $0 \leq j < M$, *then*

$$N_{\mathrm{zeros}}(\Omega) \in \mathbf{N}(\Omega) := \frac{1}{2\pi} \sum_{j=0}^{M-1} \arctan\left(\frac{R_j I_{j+1} - I_j R_{j+1}}{R_j R_{j+1} + I_j I_{j+1}} \right) = [a_\Omega, b_\Omega], \tag{4.6}$$

and $[a_\Omega, b_\Omega] \subset (-0.5, +0.5)$ *implies* $N_{\mathrm{zeros}}(\Omega) = 0$.

Proof. (a) Since $R^{(k)}(d,t) = \mathrm{Re}(\Xi^{(k)}(t - \mathrm{i}d))$ and $\partial_d \Xi^{(m)}(t - \mathrm{i}d) = -\mathrm{i}\Xi^{(m+1)}(t - \mathrm{i}d)$, $|\partial_d^p \partial_t^q R^{(k)}| \leq |\Xi^{(p+q+k)}(t - \mathrm{i}d)| \leq M_{p+q+k}$. Next, inserting $\mathrm{sinc}_{\mathrm{hyp}}(2du) = \int_0^1 \cosh(2sdu)\, \mathrm{d}s$ from (3.7) into (3.8) and applying Fubini's theorem represents $\widetilde{I}(d,t)$ directly in terms of the holomorphic derivative $\Xi'(t - \mathrm{i}sd)$:

$$\widetilde{I}(d,t) = -\int_0^1 \mathrm{Re}(\Xi'(t - \mathrm{i}sd))\, \mathrm{d}s \implies \widetilde{I}^{(k)}(d,t) = -\int_0^1 \mathrm{Re}(\Xi^{(k+1)}(t - \mathrm{i}sd))\, \mathrm{d}s.$$

Differentiating p times in d and q times in t brings down s^p inside $[0,1]$:

$$|\partial_d^p \partial_t^q \widetilde{I}^{(k)}(d,t)| \leq \int_0^1 s^p\, |\Xi^{(p+q+k+1)}(t - \mathrm{i}sd)|\, \mathrm{d}s \leq M_{p+q+k+1} \int_0^1 s^p\, \mathrm{d}s = \frac{M_{p+q+k+1}}{p+1},$$

since $sd \in [0, \overline{d}]$ whenever $d \in [\underline{d}, \overline{d}] \subset [0, \overline{d}]$. Identically, substituting the double integral for $\eta_{\mathrm{hyp}}(2du)$ gives $\widehat{I}^{(k)}(d,t) = \int_0^1 \int_0^1 s^2\, \mathrm{Re}(\Xi^{(k+3)}(t - \mathrm{i}s\tau d))\, \mathrm{d}\tau\, \mathrm{d}s$, so differentiating p times in d and q times in t yields

$$|\partial_d^p \partial_t^q \widehat{I}^{(k)}(d,t)| \leq M_{p+q+k+3} \int_0^1 s^{p+2}\, \mathrm{d}s \int_0^1 \tau^p\, \mathrm{d}\tau = \frac{M_{p+q+k+3}}{(p+3)(p+1)},$$

proving (4.4). In `scripts/verify_paper2_parity_reduced_wronskian.py`, M_m is enclosed rigorously on $[0, \overline{d}] \times [\underline{t}, \overline{t}]$ by expanding $\Xi^{(m)}(t - \mathrm{i}d)$ in t around t_0 up to order $K = 12$ with maximum-modulus bound on $\overline{d}$. Applying bivariate Taylor's theorem to each F and the mean-value theorem to $\widetilde{W}$ gives (4.5). Part (b) follows from $\mathrm{Re}(\overline{\Xi(z_j)}\Xi(z)) \geq |\Xi(z_j)|(|\Xi(z_j)| - h_j M_{1,j}) > 0$ on $[z_j, z_{j+1}]$, confining $\Xi([z_j, z_{j+1}])$ to an open half-plane so the principal branch of arctan captures the exact winding increment. □

5. Methodological interval benchmarks and IEEE-754 cancellation audit

All computations in this section are executed by `scripts/verify_paper2_parity_reduced_wronskian.py` at working precisions `mp.dps = 50` and `iv.dps = 35`. Because the Riemann Hypothesis is already verified up to $T = 3 \cdot 10^{12}$ [4, 20, 23, 13], all enclosures on $t \in [0, 55]$ serve strictly as validation benchmarks for the division-free parity-reduced interval scheme.

5.1. Interval enclosures of $c_1(\gamma_k), c_3(\gamma_k), c_5(\gamma_k) > 0$ at $\gamma_1, \dots, \gamma_{10}$.

At $(d,t) = (0.20, \gamma_1)$, the 50-digit residuals between the canonical Wigner integrals (1.3), (2.2), (3.2) and the Cauchy–Riemann expressions are $|W_{\mathrm{Wig}} - W_{\mathrm{CR}}| = 8.23 \times 10^{-55}$, $|\widetilde{W}_{\mathrm{Wig}} - \widetilde{W}_{\mathrm{CR}}| = 4.57 \times 10^{-54}$, $|c_{1,\mathrm{Wig}} - L_1(\Xi)| = 4.53 \times 10^{-54}$, $|c_{3,\mathrm{Wig}} - 2L_2(\Xi)| = 1.19 \times 10^{-55}$, and $|c_{5,\mathrm{Wig}} - 3L_3(\Xi)| = 2.27 \times 10^{-57}$.

For each of the first ten Riemann zeros $\gamma_1, \dots, \gamma_{10}$, we bracket the exact root inside $\boldsymbol{\gamma}_k = [\underline{\gamma}_k, \overline{\gamma}_k]$ of radius 10^{-16} via $\Xi(\underline{\gamma}_k)\Xi(\overline{\gamma}_k) < 0$ in `mpmath.iv` and enclose $c_1(\boldsymbol{\gamma}_k), c_3(\boldsymbol{\gamma}_k), c_5(\boldsymbol{\gamma}_k)$ via (3.3)–(3.5) (Table 2). All ten zeros satisfy $c_1(\boldsymbol{\gamma}_k).a, c_3(\boldsymbol{\gamma}_k).a, c_5(\boldsymbol{\gamma}_k).a > 0$, with local convexity ratio $c_3(\gamma_k)/c_1(\gamma_k)$ rising from 0.1354 at γ_1 to 1.1022 at γ_{10}.

5.2. Dual evaluator cross-check, 80 collar boxes, and 5 Cauchy rectangles.

Table 3 reports the three interval benchmark tiers from Section 4:

(1) *Part A (Dual evaluator overlap):* Across seven test points spanning $d \in [0, 0.50]$ and $t \in [14.13, 49.77]$ (including mid-strip evaluations at the true Ξ'-extrema $\tau_1 = 15.58570859$ and $\tau_2 = 22.09797728$), Evaluator A and Evaluator B overlap in both real and imaginary parts with $\widetilde{W}$ widths between 4.99×10^{-38} and 1.38×10^{-26}.

TABLE 2. Directed-rounding interval enclosures (`mpmath.iv`) of $\Xi(\underline{\gamma}_k)\Xi(\overline{\gamma}_k) < 0$ and lower endpoints $.a > 0$ of the Jensen–Laguerre moments $c_1 = L_1$, $c_3 = 2L_2$, $c_5 = 3L_3$ at $\gamma_1, \ldots, \gamma_{10}$.

Zero	Height γ_k	IVT Root Sign	$c_1(\boldsymbol{\gamma}_k).a = L_1$	$c_3(\boldsymbol{\gamma}_k).a = 2L_2$	$c_5(\boldsymbol{\gamma}_k).a = 3L_3$	c_3/c_1
γ_1	14.13472514	$\Xi(\underline{\gamma}_1)\Xi(\overline{\gamma}_1) < 0$	$+1.9119120797 \times 10^{-6}$	$+2.5880680426 \times 10^{-7}$	$+1.1553062387 \times 10^{-8}$	0.1354
γ_2	21.02203964	$\Xi(\underline{\gamma}_2)\Xi(\overline{\gamma}_2) < 0$	$+3.1505455533 \times 10^{-10}$	$+8.7511716020 \times 10^{-11}$	$+6.9388971291 \times 10^{-12}$	0.2778
γ_3	25.01085758	$\Xi(\underline{\gamma}_3)\Xi(\overline{\gamma}_3) < 0$	$+1.6007403799 \times 10^{-12}$	$+5.2814551425 \times 10^{-13}$	$+5.2096732221 \times 10^{-14}$	0.3299
γ_4	30.42487613	$\Xi(\underline{\gamma}_4)\Xi(\overline{\gamma}_4) < 0$	$+5.8163397852 \times 10^{-16}$	$+3.2327969937 \times 10^{-16}$	$+4.3804994322 \times 10^{-17}$	0.5558
γ_5	32.93506159	$\Xi(\underline{\gamma}_5)\Xi(\overline{\gamma}_5) < 0$	$+1.6721352814 \times 10^{-17}$	$+9.8688801466 \times 10^{-18}$	$+1.4814763006 \times 10^{-18}$	0.5902
γ_6	37.58617816	$\Xi(\underline{\gamma}_6)\Xi(\overline{\gamma}_6) < 0$	$+3.4996421209 \times 10^{-20}$	$+1.7579841347 \times 10^{-20}$	$+2.6935163846 \times 10^{-21}$	0.5023
γ_7	40.91871901	$\Xi(\underline{\gamma}_7)\Xi(\overline{\gamma}_7) < 0$	$+1.4872866506 \times 10^{-22}$	$+1.1082406339 \times 10^{-22}$	$+2.2303339091 \times 10^{-23}$	0.7451
γ_8	43.32707328	$\Xi(\underline{\gamma}_8)\Xi(\overline{\gamma}_8) < 0$	$+6.2545548444 \times 10^{-24}$	$+4.3519297910 \times 10^{-24}$	$+8.1992068916 \times 10^{-25}$	0.6958
γ_9	48.00515088	$\Xi(\underline{\gamma}_9)\Xi(\overline{\gamma}_9) < 0$	$+4.2143436862 \times 10^{-27}$	$+4.3044181465 \times 10^{-27}$	$+9.7335354676 \times 10^{-28}$	1.0209
γ_{10}	49.77383248	$\Xi(\underline{\gamma}_{10})\Xi(\overline{\gamma}_{10}) < 0$	$+2.4336218792 \times 10^{-28}$	$+2.6849833622 \times 10^{-28}$	$+6.9701273267 \times 10^{-29}$	1.1022

(2) *Part B (80 non-harmonic-safe 2D Interval-MVT collar boxes):* Across five critical-line collar zones of width $d \in [0, 0.50]$ centered at $\gamma_1, \tau_1, \gamma_2, \tau_2, \gamma_3$, all 80 boxes $\mathbf{B}_m$ satisfy $\widetilde{W}(\mathbf{B}_m).a > 0$ via Proposition 4.1(a) without subdivision beyond depth 0.

(3) *Part C (Integer argument principle on $\Omega_1, \ldots, \Omega_5$):* Covering $[10^{-6}, 1.20] \times [0, 55]$ by five closed rectangles $\Omega_1, \ldots, \Omega_5$, every boundary segment satisfies the continuum half-plane conditions of Proposition 4.1(b), yielding $|N_{\text{zeros}}(\Omega_k)| \le 2.62 \times 10^{-14} < 0.5$ and hence $N_{\text{zeros}}(\Omega_k) = 0$.

TABLE 3. Methodological benchmark enclosures in `mpmath.iv` (`iv.dps = 35`): (A) dual evaluator cross-check, (B) 80 non-harmonic-safe 2D Interval-MVT collar boxes around γ_k and true Ξ'-extrema τ_k, and (C) Cauchy argument-principle rectangles $\Omega_1, \ldots, \Omega_5$.

Region	Domain	Rigorous Enclosure	Width / Metric	Status
Part A: Dual Evaluator Cross-Check (Euler–Maclaurin A vs. Jacobi-Theta B)				
Pt 1 (γ_1, line)	$(0.00, 14.1347)$	$\widetilde{W}.a = +1.911912 \times 10^{-6}$	$\Delta\widetilde{W} = 4.99 \times 10^{-38}$	A ∩ B ≠ ∅, $\widetilde{W} > 0$
Pt 2 (γ_1, collar)	$(0.10, 14.1347)$	$\widetilde{W}.a = +1.914501 \times 10^{-6}$	$\Delta\widetilde{W} = 7.05 \times 10^{-29}$	A ∩ B ≠ ∅, $\widetilde{W} > 0$
Pt 3 (τ_1, mid-strip)	$(0.25, 15.5857)$	$\widetilde{W}.a = +3.633605 \times 10^{-7}$	$\Delta\widetilde{W} = 1.38 \times 10^{-26}$	A ∩ B ≠ ∅, $\widetilde{W} > 0$
Pt 4 (γ_2, collar)	$(0.35, 21.0220)$	$\widetilde{W}.a = +3.258794 \times 10^{-10}$	$\Delta\widetilde{W} = 5.46 \times 10^{-28}$	A ∩ B ≠ ∅, $\widetilde{W} > 0$
Pt 5 (τ_2, mid-strip)	$(0.25, 22.0980)$	$\widetilde{W}.a = +6.505722 \times 10^{-11}$	$\Delta\widetilde{W} = 1.73 \times 10^{-28}$	A ∩ B ≠ ∅, $\widetilde{W} > 0$
Pt 6 (γ_3, edge)	$(0.50, 25.0109)$	$\widetilde{W}.a = +1.736072 \times 10^{-12}$	$\Delta\widetilde{W} = 5.00 \times 10^{-29}$	A ∩ B ≠ ∅, $\widetilde{W} > 0$
Pt 7 (γ_{10}, outer)	$(0.45, 49.7738)$	$\widetilde{W}.a = +3.007916 \times 10^{-28}$	$\Delta\widetilde{W} = 5.09 \times 10^{-37}$	A ∩ B ≠ ∅, $\widetilde{W} > 0$
Part B: Non-Harmonic-Safe 2D Interval-MVT Enclosures Across 80 Collar Boxes ($d \in [0, 0.50]$)				
Zone 1 (γ_1)	$[0, 0.50] \times [14.115, 14.155]$	$\widetilde{W}(\mathbf{B}).a \ge +1.8651 \times 10^{-6}$	$\min \widehat{W}.a = +8.29 \times 10^{3}$	$16/16 > 0$
Zone 2 (τ_1)	$[0, 0.50] \times [15.565, 15.605]$	$\widetilde{W}(\mathbf{B}).a \ge +3.5158 \times 10^{-7}$	$\min \widehat{W}.a = +1.52 \times 10^{4}$	$16/16 > 0$
Zone 3 (γ_2)	$[0, 0.50] \times [21.002, 21.042]$	$\widetilde{W}(\mathbf{B}).a \ge +3.0492 \times 10^{-10}$	$\min \widehat{W}.a = +6.83 \times 10^{4}$	$16/16 > 0$
Zone 4 (τ_2)	$[0, 0.50] \times [22.078, 22.118]$	$\widetilde{W}(\mathbf{B}).a \ge +6.1711 \times 10^{-11}$	$\min \widehat{W}.a = +7.48 \times 10^{4}$	$16/16 > 0$
Zone 5 (γ_3)	$[0, 0.50] \times [24.991, 25.031]$	$\widetilde{W}(\mathbf{B}).a \ge +1.5592 \times 10^{-12}$	$\min \widehat{W}.a = +1.82 \times 10^{5}$	$16/16 > 0$
Part C: Integer Argument Principle Winding Enclosures $N_{\text{zeros}}(\Omega_k) \in [a_k, b_k] \cap \mathbb{Z}$				
Box Ω_1	$[10^{-6}, 1.20] \times [0.0, 12.0]$	$\lvert N_{\text{zeros}}(\Omega_1)\rvert \le 9.07 \times 10^{-33}$	$\min \mu_j = +7.91 \times 10^{-3}$	$N_{\text{zeros}} = 0$
Box Ω_2	$[10^{-6}, 1.20] \times [12.0, 23.0]$	$\lvert N_{\text{zeros}}(\Omega_2)\rvert \le 2.27 \times 10^{-24}$	$\min \mu_j = +1.51 \times 10^{-11}$	$N_{\text{zeros}} = 0$
Box Ω_3	$[10^{-6}, 1.20] \times [23.0, 35.0]$	$\lvert N_{\text{zeros}}(\Omega_3)\rvert \le 1.32 \times 10^{-20}$	$\min \mu_j = +3.48 \times 10^{-15}$	$N_{\text{zeros}} = 0$
Box Ω_4	$[10^{-6}, 1.20] \times [35.0, 45.0]$	$\lvert N_{\text{zeros}}(\Omega_4)\rvert \le 2.41 \times 10^{-17}$	$\min \mu_j = +2.13 \times 10^{-18}$	$N_{\text{zeros}} = 0$
Box Ω_5	$[10^{-6}, 1.20] \times [45.0, 55.0]$	$\lvert N_{\text{zeros}}(\Omega_5)\rvert \le 2.62 \times 10^{-14}$	$\min \mu_j = +2.05 \times 10^{-21}$	$N_{\text{zeros}} = 0$

5.3. **IEEE-754 `float64` catastrophic cancellation audit.** It is instructive to ask whether fast GPU IEEE-754 double-precision (`float64`, $u_{\text{mach}} = 2^{-53} \approx 1.11 \times 10^{-16}$) quadrature of (3.8) combined with heuristic vertex-gradient margins can certify $\widehat{W}(d, t) = \mathrm{e}^{\pi t/2}\widetilde{W}(d, t) > 0$. The answer is decisively negative for two independent reasons: (i) multiplying a discrete four-corner gradient maximum by an empirical factor (such as 1.05) provides no mathematical upper bound on

inter-node second derivatives; and (ii) Higham's inner-product rounding theorem [6, Theorem 3.1] bounds the floating-point error of an oscillatory quadrature sum $x^\top y = \sum_{j=0}^{N_u} w_j \cos(2tu_j)$ in terms of the *unoscillatory absolute inner product* $|x|^\top|y| = \mathcal{O}(1)$, not the exponentially cancelled value $|x^\top y| \asymp t^{7/4}\mathrm{e}^{-\pi t/4}$:

$$|\,\mathrm{fl}(x^\top y) - x^\top y| \le \gamma_{N_u}\,|x|^\top|y| = \mathcal{O}(\gamma_{N_u}), \qquad \gamma_{N_u} := \frac{N_u\,u_{\mathrm{mach}}}{1 - N_u\,u_{\mathrm{mach}}}. \tag{5.1}$$

Because each factor in $\widetilde{W} = R\widetilde{I}_t - \widetilde{I}R_t$ has exact magnitude $\mathcal{O}(t^\alpha \mathrm{e}^{-\pi t/4})$ while its `float64` evaluation carries additive absolute noise $\mathcal{O}(u_{\mathrm{mach}})$, multiplying $\widetilde{W}$ by $\mathrm{e}^{\pi t/2}$ amplifies the bilinear roundoff error exponentially:

$$|\widehat{W}_{\mathrm{float64}}(d,t) - \widehat{W}(d,t)| \lesssim \gamma_{N_u}\Big(\|w_R\|_1|\widetilde{I}_t| + |R|\|w_{\widetilde{I}_t}\|_1 + \|w_{\widetilde{I}}\|_1|R_t| + |\widetilde{I}|\|w_{R_t}\|_1\Big)\mathrm{e}^{\pi t/2} \propto u_{\mathrm{mach}}\,\mathrm{e}^{+\pi t/4}. \tag{5.2}$$

Table 4 and Figure 2(c) verify (5.2) directly against 50-digit `mpmath` at $d = 0.05$ ($N_u = 4096$): while `float64` is accurate for plotting on $t \le 45$ ($\le 2\times10^{-5}$ relative error), by $t = 55$ its absolute error reaches 5.15×10^5 (4.51% relative error), and by $t = 60$ ($\mathrm{e}^{\pi\cdot 60/4} \approx 2.9 \times 10^{20} \gg u_{\mathrm{mach}}^{-1}$) relative error explodes to 1,375%. Rigorous enclosures via direct Fourier–Jacobi quadrature therefore require arbitrary-precision interval arithmetic whose working precision scales linearly with height, $\mathrm{dps}(T) > \frac{\pi T}{4\ln 10}$.

TABLE 4. IEEE-754 `float64` cancellation-amplification audit ($N_u = 4096$, $d = 0.05$) versus 50-digit `mpmath` and Higham's $|x|^\top|y|$ bound (5.2).

Height t	$\widehat{W}_{\mathrm{exact}}$ **(50 dps)**	$\widehat{W}_{\mathrm{float64}}$	$\lvert\Delta\widehat{W}\rvert$ **(Measured)**	**Rel. Error**	**Higham** $\lvert x\rvert^\top\lvert y\rvert$ **Bound**
10.0	$+1.01635042 \times 10^3$	$+1.01635042 \times 10^3$	2.3874×10^{-12}	2.35×10^{-15}	2.8163×10^{-8}
20.0	$+6.37433265 \times 10^4$	$+6.37433265 \times 10^4$	1.9456×10^{-8}	3.05×10^{-13}	7.6427×10^{-4}
30.0	$+3.81160142 \times 10^5$	$+3.81160143 \times 10^5$	3.9555×10^{-4}	1.04×10^{-9}	5.5699×10^{0}
35.0	$+1.36413722 \times 10^6$	$+1.36413723 \times 10^6$	7.1859×10^{-3}	5.27×10^{-9}	1.5172×10^{2}
40.0	$+1.35386772 \times 10^6$	$+1.35387006 \times 10^6$	2.3401×10^{0}	1.73×10^{-6}	1.5487×10^{4}
45.0	$+5.45986814 \times 10^6$	$+5.45975732 \times 10^6$	1.1082×10^{2}	2.03×10^{-5}	1.0239×10^{6}
50.0	$+2.58544099 \times 10^6$	$+2.57931492 \times 10^6$	6.1261×10^{3}	2.37×10^{-3}	5.8917×10^{7}
55.0	$+1.14236179 \times 10^7$	$+1.09090204 \times 10^7$	5.1460×10^{5}	4.51×10^{-2}	2.2602×10^{9}
60.0	$+2.93092890 \times 10^6$	$+4.32403216 \times 10^7$	4.0309×10^{7}	1.38×10^{1}	2.3009×10^{11}

6. Scope, limitations, and explicit obstructions

We record the exact scope of the identities proved above and the structural obstructions separating finite-height interval benchmarks from asymptotic zero-free results:

(1) **Davenport–Heilbronn and explicit off-critical zero counterexample ($L_1 < 0$):** Proposition 2.1, Theorem 3.1, and Proposition 3.2 use only Schwarz reflection $F(\bar{z}) = \overline{F(z)}$ and entireness (or Fourier decay), never the Euler product of $\zeta(s)$. Consequently, parity reduction $\widetilde{W}_F(d,t) = W_F(d,t)/d$ and $c_{2m-1}(F;t) = m\,L_m(F;t)$ hold verbatim for the Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$ [3], which satisfies the same Γ-functional equation as Ξ yet has infinitely many zeros in $d > 0$, as well as for the explicit real-entire perturbation

$$F_{\mathrm{off}}(z) := \Xi(z)\,\frac{(z^2 - z_*^2)(z^2 - \bar{z}_*^2)}{(z^2 - \gamma_1^2)(z^2 - \gamma_2^2)}, \qquad z_* = t_* - \mathrm{i}d_*, \quad t_* = \tfrac{1}{2}(\gamma_1 + \gamma_2) \approx 17.5784, \quad d_* = 0.15.$$

As verified to 50 decimal digits in the companion script (`scripts/verify_paper2_parity_reduced_wronskian.py`), $\widetilde{W}_{F_{\mathrm{off}}}(0,t_*) = L_1(F_{\mathrm{off}};t_*) = -5.095559 \times 10^{-11} < 0$. By Csordas and Varga [2], proving $c_{2m-1}(t) = m\,L_m(\Xi;t) > 0$ globally for all $m \ge 1$ and $t \in \mathbb{R}$ is equivalent to RH, not an unconditional consequence of the $\mathrm{sinc}_{\mathrm{hyp}}(2dy) \ge 1$ representation (since $I(y,t)$ oscillates in sign [11]).

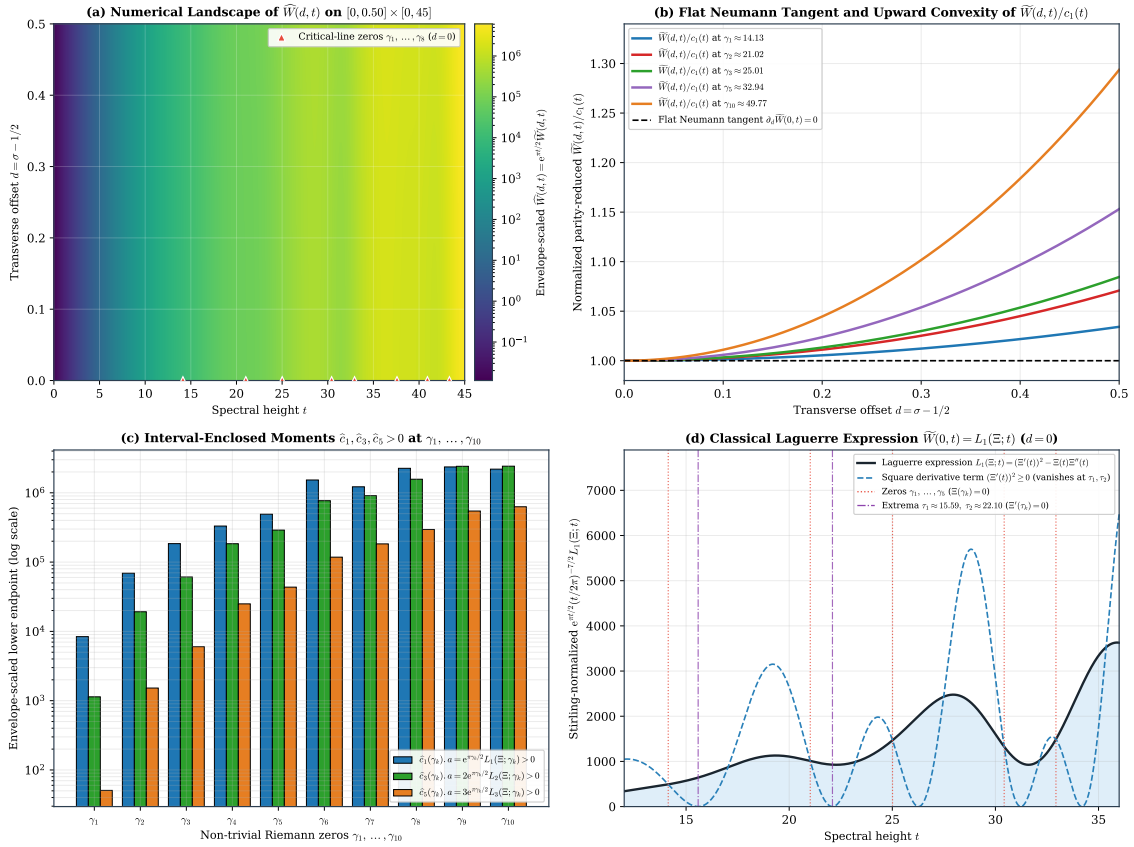


FIGURE 1. (a) Exploratory numerical landscape of $\widehat{W}(d,t)=\mathrm{e}^{\pi t/2}\widetilde{W}(d,t)$ on $[0,0.50]\times[0,45]$ alongside critical-line zeros $\gamma_1,\ldots,\gamma_8$. (b) Normalized profiles $\widetilde{W}(d,t)/c_1(t)$ across $d\in[0,0.50]$ showing the flat Neumann tangent $\partial_d\widetilde{W}(0,t)=0$. (c) Interval-enclosed moments $\widehat{c}_1(\boldsymbol{\gamma}_k).a,\widehat{c}_3(\boldsymbol{\gamma}_k).a,\widehat{c}_5(\boldsymbol{\gamma}_k).a>0$ at $\gamma_1,\ldots,\gamma_{10}$. (d) Classical Laguerre expression $\widetilde{W}(0,t)=L_1(\Xi;t)$ versus $(\Xi'(t))^2$ on $t\in[12,36]$, highlighting zeros γ_k and true Ξ'-extrema $\tau_1\approx15.59$, $\tau_2\approx22.10$.

(2) **Lehmer phenomenon and Skewes-scale Dirichlet phase alignment:** At critical-line extrema $\Xi'(\tau_k)=0$, $L_1(\Xi;\tau_k)=-\Xi(\tau_k)\Xi''(\tau_k)$ can become arbitrarily small whenever two zeros form a close Lehmer pair [10], requiring correspondingly finer interval boxes Δt as logarithmic Dirichlet phases $\cos(t\log p)$ align coherently at large t.

(3) **Voronin universality off the critical line:** By Voronin's universality theorem [24], on $\frac{1}{2}<\sigma<1$ $(0<d<\frac{1}{2})$ the shifts $\zeta(\sigma+\mathrm{i}(t+\tau))$ approximate arbitrary non-vanishing holomorphic functions, ruling out any purely local Archimedean lower bound on $\widetilde{W}(d,t)$ at large t that ignores arithmetic cancellation.

(4) **Exponential quadrature precision scaling** $(\mathrm{dps}(T)\gtrsim\frac{\pi T}{4\ln 10})$**:** As quantified in Table 4, direct Fourier–Jacobi quadrature of (1.1) cancels $\approx\frac{\pi t}{4\ln 10}\approx0.341\,t$ decimal digits, making large-height verification via raw Φ-quadrature computationally prohibitive compared to FFT-accelerated Riemann–Siegel interval methods [12, 13].

APPENDIX A. EXPLICIT ANALYTICAL ERROR BOUNDS FOR SERIES, DOMAIN, AND QUADRATURE TRUNCATION

Lemma A.1 (Rigorous truncation bounds for interval evaluators)**.** *Let* $(d,t)\in[0,1.20]\times[0,55]$. *Then:*

(a) ***Power-series remainders near*** $w=0$***:*** *For* $0\leq w\leq0.10$, *truncating the Maclaurin series of* $\cosh(w)$, $\mathrm{sinc}_{\mathrm{hyp}}(w)$, *and* $\eta_{\mathrm{hyp}}(w)$ *after* w^{10} *(*$k=5$*) leaves non-negative remainders bounded by* $1.01\frac{w^{12}}{12!}$, $1.01\frac{w^{12}}{13!}$, *and* $1.01\frac{14\,w^{12}}{15!}$, *respectively.*

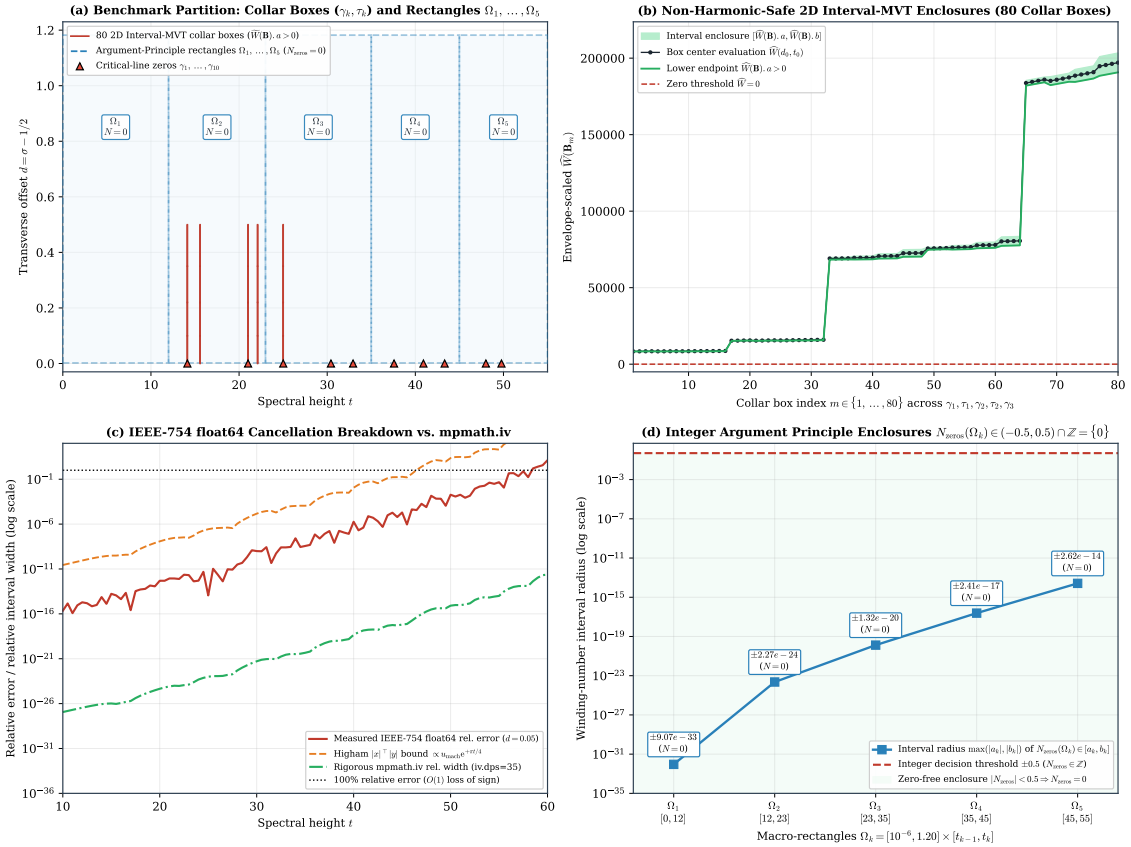


FIGURE 2. (a) Benchmark partition on $[0, 55]$: 80 2D Interval-MVT collar boxes $\mathbf{B}_m$ around γ_k, τ_k and five Cauchy rectangles $\Omega_1, \ldots, \Omega_5$. (b) Non-harmonic-safe 2D Interval-MVT enclosures $[\widehat{W}(\mathbf{B}_m).a, \widehat{W}(\mathbf{B}_m).b]$ across Zones 1–5. (c) Exponential IEEE-754 `float64` relative error growth ($\propto u_{\text{mach}} e^{+\pi t/4}$) versus Higham's $|x|^\top |y|$ bound and `mpmath.iv` (`iv.dps=35`) relative width across $t \in [10, 60]$. (d) Argument-principle winding radii $\max(|a_k|, |b_k|) \le 2.62 \times 10^{-14} < 0.5$ proving $N_{\text{zeros}}(\Omega_k) = 0$.

(b) ***Euler–Maclaurin remainder*** (4.2)***:*** *For* $\sigma = \frac{1}{2} + d > 0$ *and integers* $N, M \ge 1$, $|R_{M,N}(s)| \le \varepsilon_{\text{EM}}(s)$.

(c) ***Jacobi series, domain tail, and strip trapezoidal bounds*** (4.3)***:*** *For* $M_{\text{ser}} = 12$, $U_{\max} = 1.55$, *and* $h = 1.55/140$, *the truncation errors across all auxiliary integrals in* (1.1) *and* (3.8) *satisfy* $\varepsilon_{\text{ser}} \le 10^{-220}$, $\varepsilon_{\text{tail}} \le 10^{-550}$, *and* $\varepsilon_{\text{quad}} \le 10^{-55}$.

Proof. Parts (a) and (b) follow respectively from geometric majorization with term ratio $\frac{w^2}{13 \cdot 14} < 10^{-4}$ and the standard Euler–Maclaurin remainder theorem [21, Section 6.4]. For part (c), for $u \ge 0$ and $n \ge 13$, $\pi n^2 e^{4u} \ge 169\pi > 530.9$, giving $\varepsilon_{\text{ser}} \le 16\pi^2 (13^4) e^{-169\pi} < 10^{-220}$; for $u \ge 1.55$ and $d \le 1.20$, $\pi e^{4u} > 1548.8$ while $(2u)^8 \cosh(2du) e^{9u} < 4.2 \times 10^{11}$, giving $\varepsilon_{\text{tail}} < 10^{-550}$. Finally, $\Phi(u)$ is even and holomorphic throughout the open strip $|\operatorname{Im}(u)| < \frac{\pi}{8} \approx 0.3927$ (note that the strip zeros of Φ at $\pm i\tau_0 \approx \pm 0.3194i$ and $\pm 0.1278 \pm 0.3478i$ analyzed in [11] create logarithmic singularities for $\log \Phi$ in saddle-point asymptotics, not poles of the holomorphic integrand $\Phi(u)(2u)^m \cosh(2du) \cos(2tu)$). On $|\operatorname{Im}(u)| \le \eta = 0.35 < \frac{\pi}{8}$, $\cos(4\eta) \approx 0.16997 > 0$ gives double-exponential decay as $\operatorname{Re}(u) \to \pm\infty$. By [22, Theorem 5.1], $\varepsilon_{\text{quad}}(h) \le 2M(\eta) e^{2(|t|+|d|)\eta} / (e^{2\pi\eta/h} - 1)$ with $2\pi\eta/h = 2\pi(0.35)(140)/1.55 \approx 198.63$, $e^{-2\pi\eta/h} \approx 5.45 \times 10^{-87}$, $e^{2(55+1.20)(0.35)} \approx 1.22 \times 10^{17}$, and $M(0.35) \le 10^{12}$, yielding $\varepsilon_{\text{quad}}(h) < 10^{-55}$. □

Use of AI and Computational Tools

Gemini (Google DeepMind) was used interactively to assist with symbolic verification of the positive-weight integral majorants, algebraic cross-checking against the classical Jensen–Csordas–Varga identities, and implementation of the Python `mpmath.iv` interval benchmark and IEEE-754 cancellation audit (`scripts/verify_paper2_parity_reduced_wronskian.py`). All analytical proofs, error bounds, citations, and numerical outputs were independently checked and verified.

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